

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
2nd Semester of B. Pharm (New) Examination
University Theory Examination October/November, 2016
MA141.1 Advanced Mathematics

Date: 22/10/2016, Saturday Time: 10:00 a.m. to 01.00 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION I is 30 minutes. Answers to SECTION I must be written during the first 30 minutes of the examination.
3. Answers to SECTION II and SECTION III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.
6. Use of scientific non-programable calculator is allowed.

SECTION I

Q.1. Darken the circle of the correct/most suitable choice. [20]

1. 18th derivative of function $y = e^{3x}$ is

- (A) $3^{18}e^{3x}$ (B) $e^{3x} + 18$ (C) 18^3e^{3x} (D) $3e^{18x}$

2. 100th derivative of function $y = 5x^7 - x^2 - 1$ is

- (A) 0 (B) 100 (C) 1 (D) 5

3. The value of $\binom{1000}{999}$ is

- (A) 999 (B) 1000 (C) 1 (D) 10999

4. 2nd derivative of function $y = \cos x$ is

- (A) $\cos x$ (B) $-\cos x$ (C) $-\sin x$ (D) $\sin x$

5. Which of the following is true?

- (A) Every Taylor series is Macluarin series.
(B) Every Macluarin series is Taylor series.
(C) Macluarin series expansion arround 0 is Taylor series.
(D) None of the above

6. Which of the following condition is not necessary for Lagrange mean value theorem?

- (A) $f(x)$ is continuous on $[a, b]$.
(B) $f(x)$ is differentiable on (a, b) .
(C) $f(a) = f(b)$
(D) None of the above

7. $\mathcal{L}[af(t) + bg(t)] = \dots$
- (A) $aL[f(t)] + bL[g(t)]$
 (B) $bL[f(t)] - aL[g(t)]$
 (C) $aL[g(t)] + bL[f(t)]$
 (D) None of Above
8. The Laplace transform of $\sin 2t$ is
- (A) $\frac{2}{s^2 + 4}$
 (B) $\frac{1}{s^2 + 4}$
 (C) $\frac{2}{s^2 - 4}$
 (D) $\frac{1}{s^2 - 4}$
9. The inverse Laplace transform of $\frac{1}{s - 30}$ is
- (A) e^{30t}
 (B) e^{-30t}
 (C) $30e^t$
 (D) None of the above
10. The Laplace transform of $\cosh 2t$ is
- (A) $\frac{s}{s^2 - 4}$
 (B) $\frac{2s}{s^2 + 4}$
 (C) $\frac{1}{s^2 + 2}$
 (D) $\frac{1}{s - 2}$
11. The equation $M dx + N dy = 0$ is exact if and only if
- (A) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$
 (B) $\frac{\partial M}{\partial y} = 1 + \frac{\partial N}{\partial x}$
 (C) M and N are homogenous
 (D) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$
12. The Wronskian of two functions $y_1 = \cos 2x$ and $y_2 = \sin 2x$ is
- (A) 2
 (B) -2
 (C) 4
 (D) None of Above
13. The order and degree of differential equation $\frac{d^4 y}{dx^4} + y = 0$ are
- (A) order=1 and degree=4
 (B) order=4 and degree=1
 (C) order=4 and degree=4
 (D) None of Above
14. If $-1/2$ and -5 are roots of auxiliary equation of second order homogenous differential equation, then solution of that equation is
- (A) $y = C_1 e^{5x} + C_2 e^{-1/2x}$
 (B) $y = C_1 e^{-5x} + x C_2 e^{-1/2x}$
 (C) $y = C_1 e^{-1/2x} + C_2 e^{-5x}$
 (D) None of the above
15. If -3 and -3 are roots of auxiliary equation of second order homogenous differential equation, then solution of that equation is

- (A) $y = C_1 e^{-3x} + C_2 e^{-3x}$ (C) $y = C_1 e^{-3x} + x C_2 e^{-3x}$
 (B) $y = C_1 e^{-3x} + C_2 e^{3x}$ (D) None of the above
16. The particular integral of differential equation $\frac{d^2 y}{dx^2} + y = 0$ is
- (A) $y_p = C_1 e^{-x} + C_2 e^x$ (C) $y_p = C_1 \cos x + C_2 \sin x$
 (B) $y_p = C_1 e^{-2x} + C_2 e^{2x}$ (D) $y_p = 0$
17. If $1 + 2i$ and $1 - 2i$ are roots of auxiliary equation of second order homogenous differential equation, then solution of that equation is
- (A) $y = e^x (C_1 \cos 2x + C_2 \sin 2x)$ (C) $y = C_1 e^{-x} + C_2 e^{2x}$
 (B) $y = e^x (C_1 \cos 2x - C_2 \sin 2x)$ (D) None of the above
18. The Complementary function of differential equation $\frac{d^2 y}{dx^2} - y = 0$ is
- (A) $y_c = 0$
 (B) $y_c = C_1 e^{-2x} + C_2 e^{2x}$
 (C) $y_c = C_1 \cos x + C_2 \sin x$
 (D) $y_c = C_1 e^{-x} + C_2 e^x$
19. If $y(t)$ is the number of bacteria present at time t hours, then which of the following indicate growth rate of the bacteria?
- (A) $\frac{dy}{dt}$ (C) $y^2(t) = 0$
 (B) $\frac{dy}{dt} = 0$ (D) None of above
20. If T is the temperature of the body at any time t and T_A is the surrounding temperature, then by Newton's law of cooling which of the following is true?
- (A) $\frac{dT}{dt} \propto T$ (C) $\frac{dT}{dt} \propto (T - T_A)$
 (B) $\frac{dT}{dt} \propto T_A$ (D) None of above

SECTION II

Q.2. Attempt any **Two** of the following. [10]

A Find n^{th} derivative of function $y = x^2 e^{3x}$ using Leibnitz's theorem. [05]

B Bacteria multiply at the rate proportional to the number present. If the original number N doubles in three hours, find in how many hours, the numbers of the bacteria will be $4N$. [05]

C Solve the differential equation $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$. [05]

Q.3. Attempt any **Two** of the following. [10]

A Verify Lagrange Mean's mean value theorem for $f(x) = x^3 + 5$, $x \in [-1, 3]$. [05]

B Solve the differential equation $(x^2 - y^2)dx = 2xydy$. [05]

C Solve the differential equation $\frac{d^2y}{dt^2} + 16y = 0$ with initial condition $y(0) = 0$ and $y'(0) = 1$ using Laplace transform. [05]

SECTION III

Q.4. Attempt any **Four** of the following. [20]

A 1. Find Laplace transform of function $e^{2t} + 5 + \sin 3t$. [03]

2. Find inverse Laplace transform of function $\frac{s-4}{(s-4)^2+9}$ [02]

B 1. 12^{th} derivative of function $y = \log(3x + 2)$ [03]

2. Find n^{th} derivative of function $y = \frac{1}{2x+5}$. [02]

C Find Maclaurin's series for the function $\sin x$. [05]

D Solve the differential equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} + 2y = 0$ [05]

E Solve the differential equation $\frac{d^2y}{dx^2} - 2\frac{y}{x} = x^2$ [05]

F Suppose that in the course of any given year the number of cases of a disease is reduced by 20 percentage. If there are 10,000 cases today, how many years it will take to reduce the number to 1000? [05]

Q.5. Attempt any **Four** of the following. [20]

A Find inverse Laplace transform of function $\frac{1}{(s-7)(s+1)}$ [05]

B 1. Find Taylor's series expansion of $f(x) = x^3 - 2x + 4$ around $a = 2$. [03]

2. Find 6th derivative of function $y = e^{5x} \cos(3x + 2)$. [02]

C Solve the differential equation Solve: $(y^2 - 2xy) dx = (x^2 - 2xy) dy$ [05]

D Show that the value of c in the conclusion of the Cauchy's mean value theorem for $f(x) = e^x$ and $g(x) = e^{-x}$ on an interval of positive numbers $[a, b]$ is the arithmetic mean of a and b [05]

E If the temperature of a body drops from 100° to 60° in one minute when the temperature of the surrounding is 20° , what will be the temperature of the body at the end of the second minute?

F 1. Find the order and degree of following differential equations

(i) $(\frac{dy}{dx})^{\frac{1}{2}} - y = 0$ (ii) $\frac{d^5y}{dx^5} + (\frac{d^2y}{dx^2})^3 + y = \sin x$. [03]

2. Find the Wronskian of two functions $y_1 = x$ and $y_2 = x^2$. [02]
